

1. Scope

This SOP covers the drive-end bearing (DE bearing) maintenance for Pump P-101A.

2. Asset Information

Asset: Centrifugal Pump A (Pump-A / CP-A / P-101A)
Component: Drive End Bearing, DE bearing
Sensor: TE-101A-DE bearing temperature RTD, sampling 1 Hz, unit Celsius.
Sensor: VE-101A-DE vibration accelerometer, metric vibration_rms, unit MM_S.

3. Failure Modes

Bearing overheat — severity DEGRADED — mechanisms OVERHEATING, WEAR
Detection metric: bearing_temp
Root cause: under lubrication / lack of grease

4. Mitigation

Execute SOP-114 Bearing Lubrication and Inspection (Rev C).
Required tooling: TORQUE-WRENCH-50NM calibrated torque wrench.
Step 1 — Isolate pump — SafetyCheck — Hold Point: true
Expected outcome: pump isolated, LOTO applied.

5. Maintenance Check Table

Check Item	Limit	Unit	Action
Bearing temperature	< 85	Celsius	Inspect lubrication
Vibration RMS	< 4.5	mm/s	Balance check
Lubrication interval	720	hours	Re-grease

End of excerpt.